

Labwork 3: Logistic Regression

Tran Ngoc Thanh Binh

May 2026

1 Implementation

The model uses *Experience* and *Salary* to predict *Loan*. The linear model is:

$$z = w_1x_1 + w_2x_2 + b$$

The prediction is calculated with the sigmoid function:

$$\hat{y} = \frac{1}{1 + e^{-z}}$$

The loss function is binary cross entropy:

$$J = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

During gradient descent, the error is:

$$error = \hat{y} - y$$

The program updates w_1 , w_2 , and b for 10000 epochs using a learning rate of 0.0001. It also plots the decision boundary:

$$w_1x_1 + w_2x_2 + b = 0$$

2 Conclusion

Logistic regression can classify loan results by converting a linear output into a probability with sigmoid. Binary cross entropy and gradient descent are used to train the weights and bias.

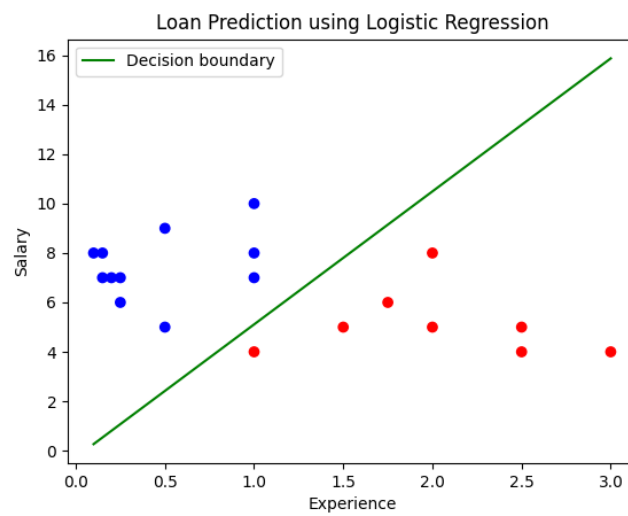


Figure 1: Loan prediction result with logistic regression decision boundary.